GI SECRETION

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GI secretion

- Salivary secretion
- Gastric secretion
- Small bowel secretion
- Large bowel secretion
- Pancreatic secretion
- Bile secretion

Daily output of GI secretion

Table 65-1 Daily Secretion of Intestinal Juices

Type of Secretion	Daily Volume (ml)	pH
Saliva	1000	6.0-7.0
Gastric secretion	1500	1.0-3.5
Pancreatic secretion	1000	8.0-8.3
Bile	1000	7.8
Small intestine secretion	1800	7.5-8.0
Brunner's gland secretion	200	8.0-8.9
Large intestinal secretion	200	7.5-8.0
Total	6700	

SALIVARY SECRETION

Function of saliva

- Lubrication (by mucus)
- Facilitation of taste
- Protection (by dilution and buffering ingested food)
- Initial digestion of starch (by amylase) and lipid (lipase)
- Cleansing the mouth and selective anti-bacterial action (lysozyme, IgA)
- Neutralization of gastric reflux
- Aid in speech

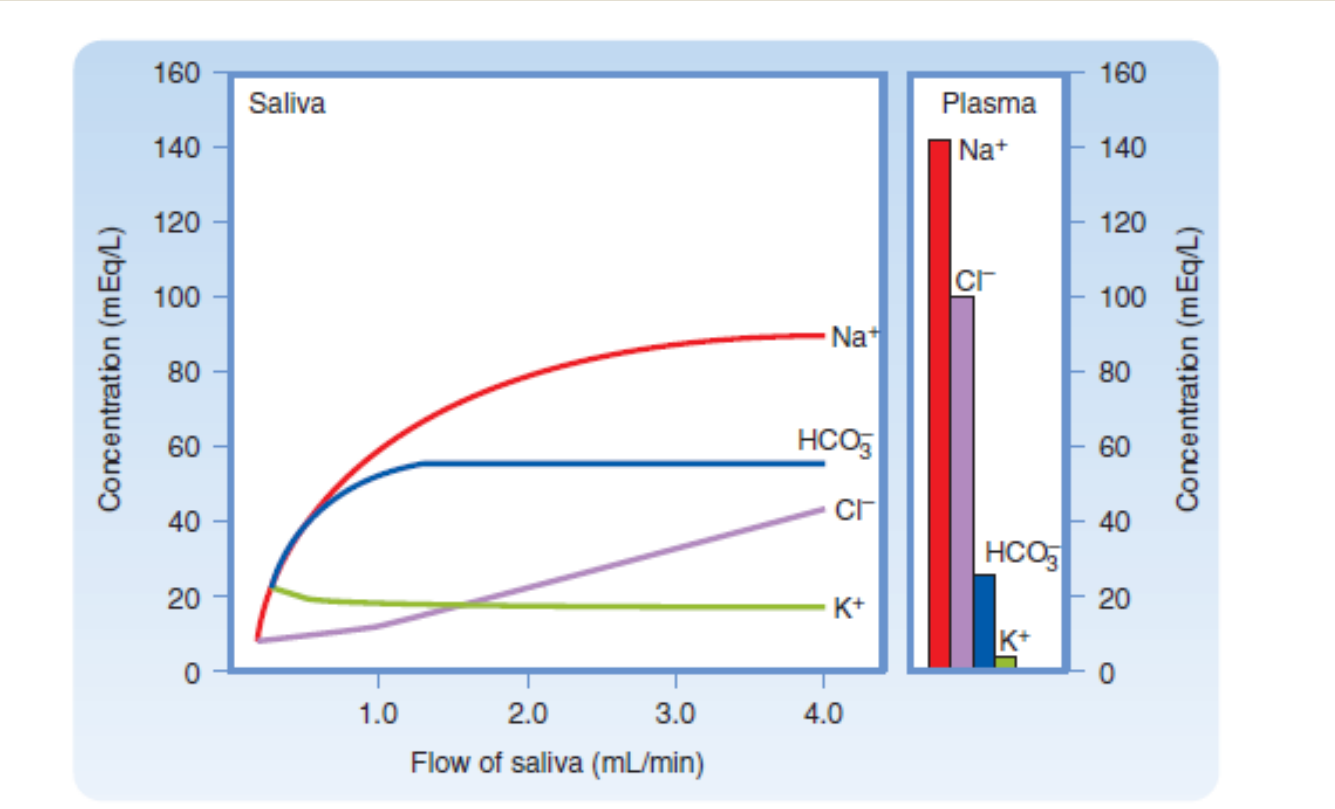
Salivary secretion

- **Primary secretion** (from acini): contains amylase and/or mucin, isotonic, similar ion concentration to plasma
- **Secondary secretion** (modified by duct cells): hypotonic, slightly alkaline, higher K than plasma

Composition of saliva

- High K and HCO_3
- Low Na and Cl
- Hypotonicity
- Alpha amylase, lingual lipase

Effects of flow rate on ions



Higher flow → more isotonic, higher Na/Cl, lower K, high HCO3
Noted: HCO3 secretion can be stimulated along with flow

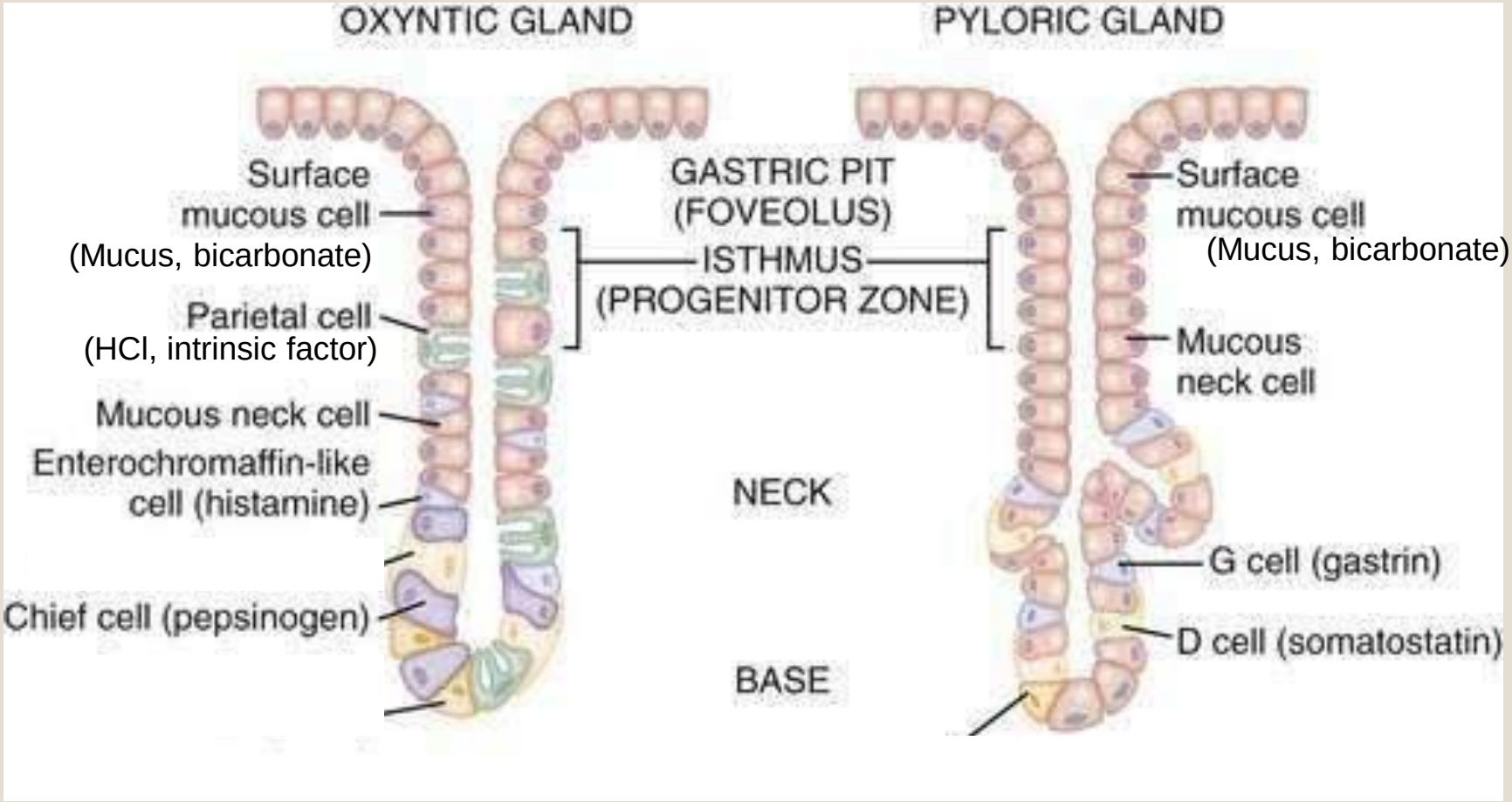
Control of salivary secretion

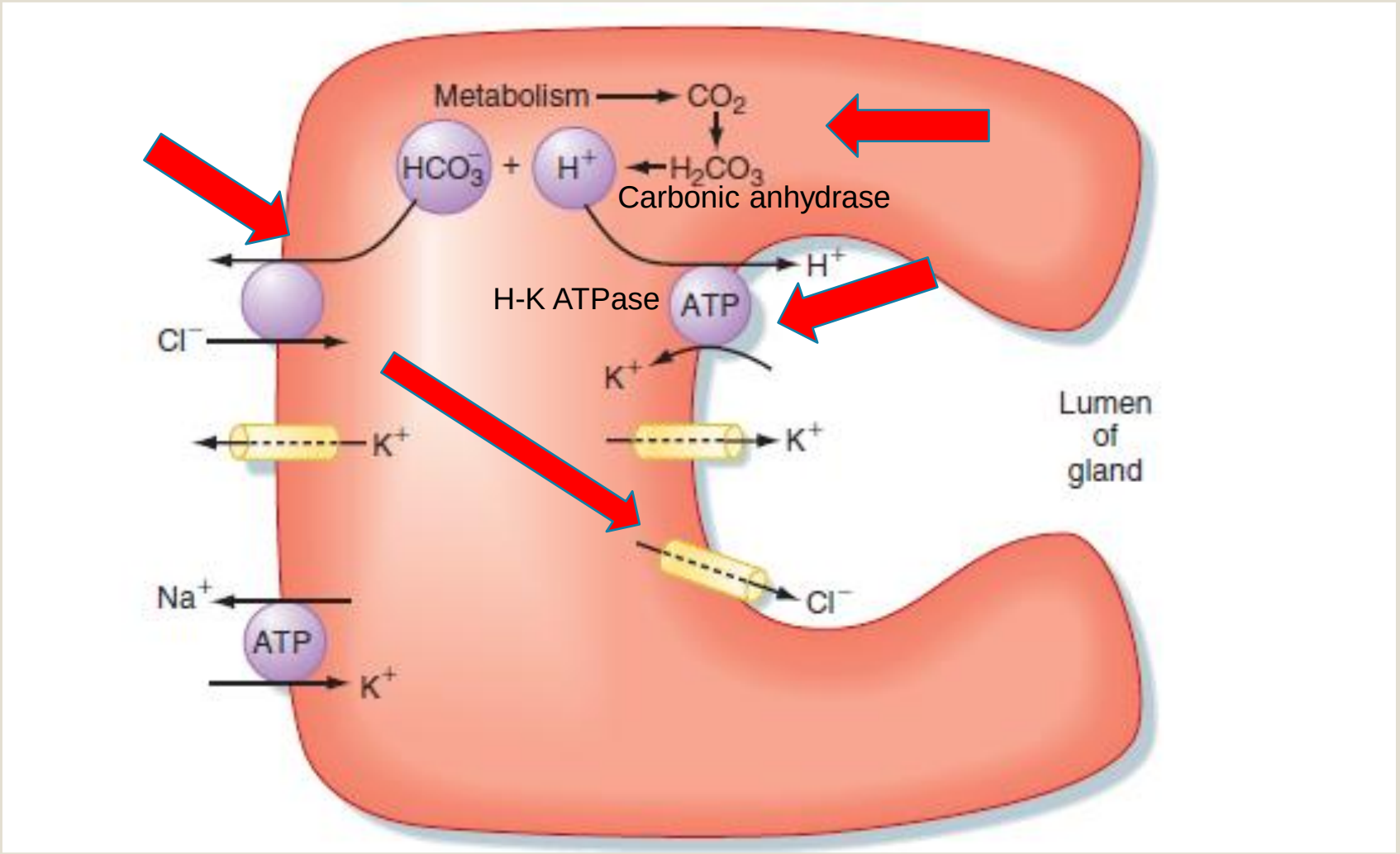
- Controlled mainly by parasympathetic from superior and inferior salivary nuclei
- **Stimuli**:
- Taste (especially sour taste) and tactile stimuli (especially smooth object) in the mouth/pharynx,
- Higher center (smell, sight)
- GI irritation, nausea
- Sympathetic stimulation: ↑proteinaceous content, small effect on volume

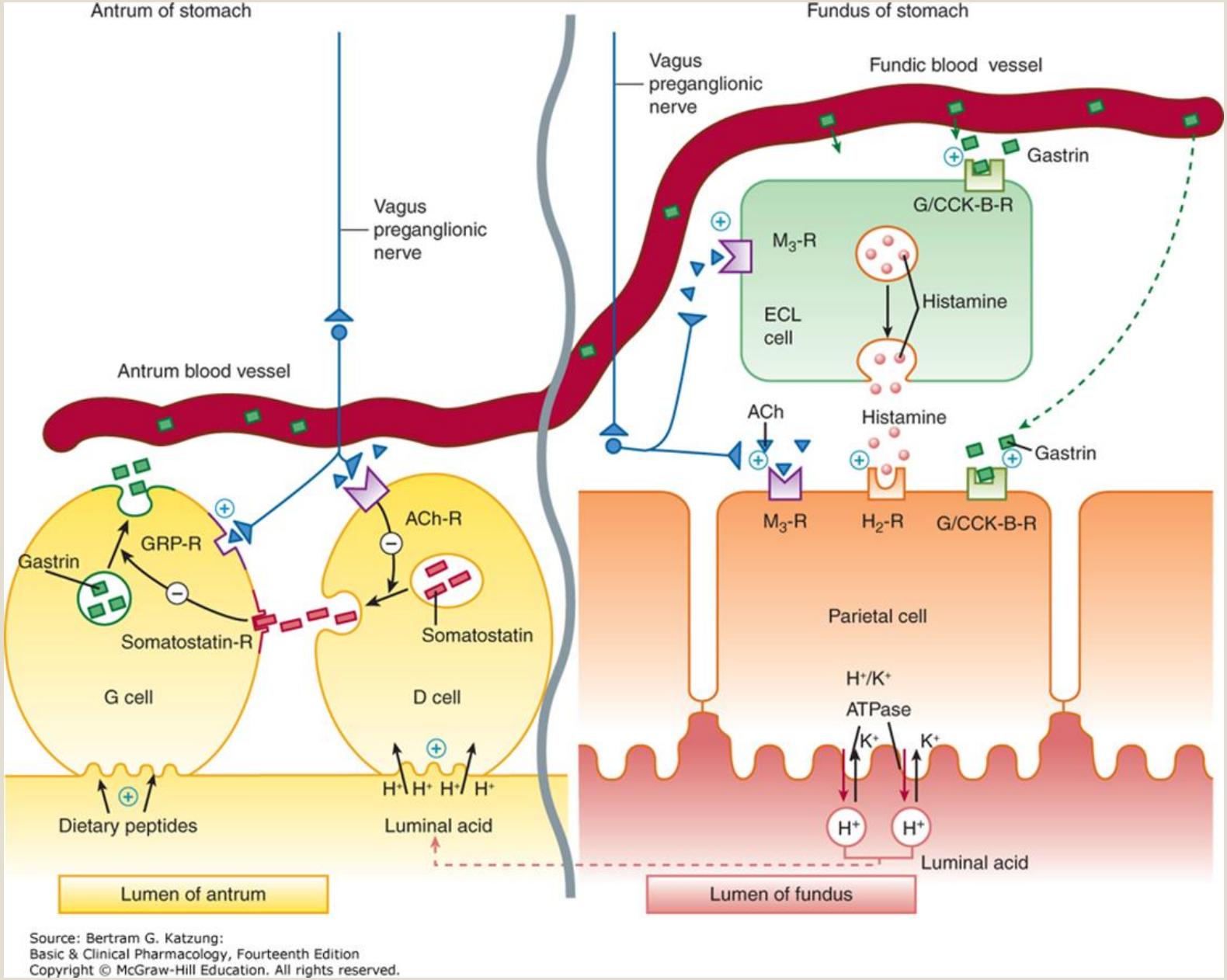
GASTRIC SECRETION

Component of gastric secretion

- Isotonic
- Higher concentration of H^+ K^+ and Cl than plasma
- Acidic
- Contains: pepsin, gastric lipase, intrinsic factor

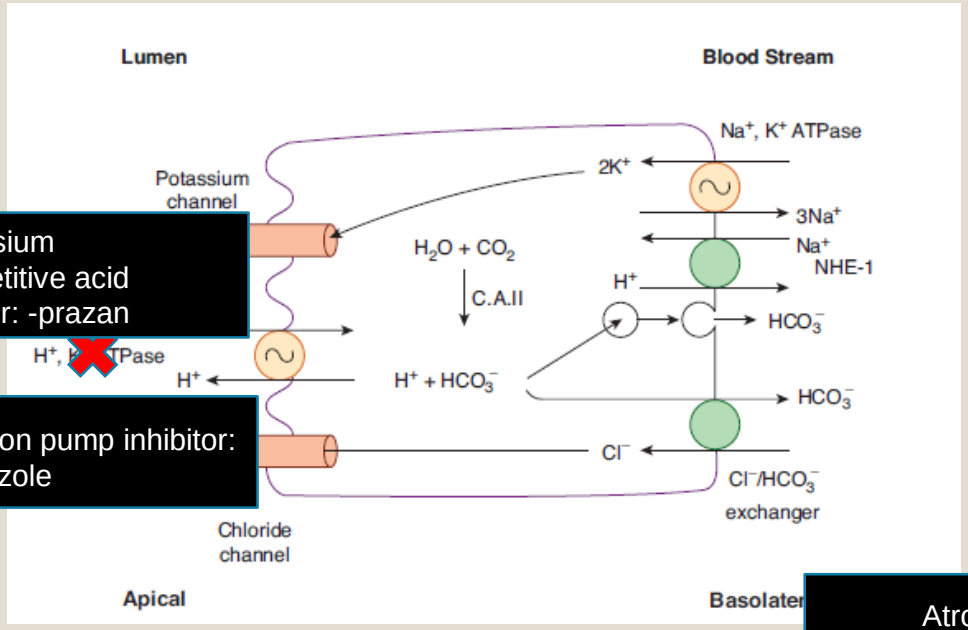

OXYNTIC GLAND
PYLORIC GLAND
Surface mucous cell (Mucus, bicarbonate)
Parietal cell (HCl, intrinsic factor)
Mucous neck cell
Enterochromaffin-like cell (histamine)
Chief cell (pepsinogen)
GASTRIC PIT (FOVEOLUS)
ISTHMUS (PROGENITOR ZONE)
NECK
BASE
Surface mucous cell (Mucus, bicarbonate)
Mucous neck cell
G cell (gastrin)
D cell (somatostatin)


Metabolism
CO_2
HCO_3^-
+
H^+
H_2CO_3
Carbonic anhydrase
H^+
H-K ATPase
ATP
Cl^-
K^+
K^+
K^+
Lumen
of
gland
Cl^-
Na^+
ATP
K^+



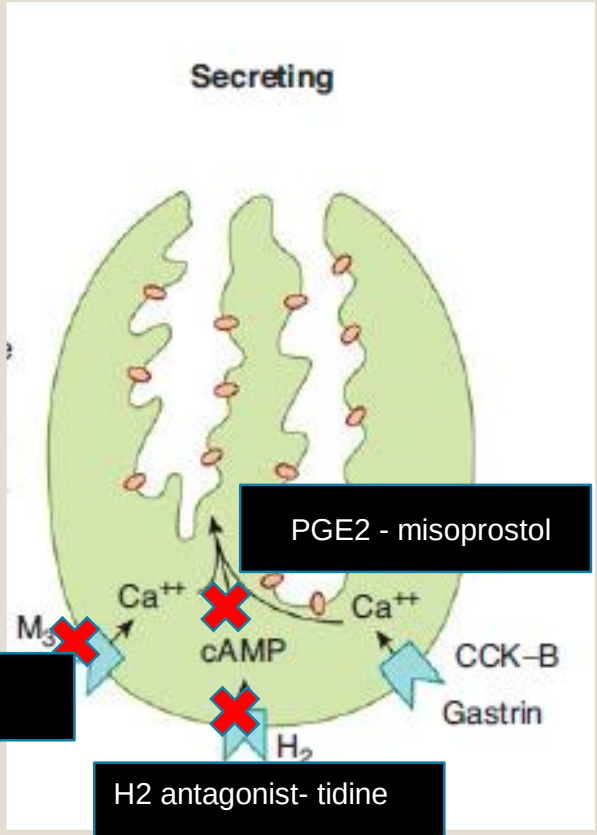
Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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Medications to control acid secretion



Potassium competitive acid blocker: -prazan

Proton pump inhibitor: -prazole



PGE2 - misoprostol

Atropine

H2 antagonist- tidine

Gastric enzymes

- **Pepsinogen** : PG I , PG II, activated by HCl to form pepsin. Functions at pH 1.8-3.5, inactivated if pH>5

  digest 20% of meal protein (not crucial because pancreatic protease can do this too)
- Gastric lipase
- Rennin : digest casein in milk
- Gelatinase
- Gastric amylase

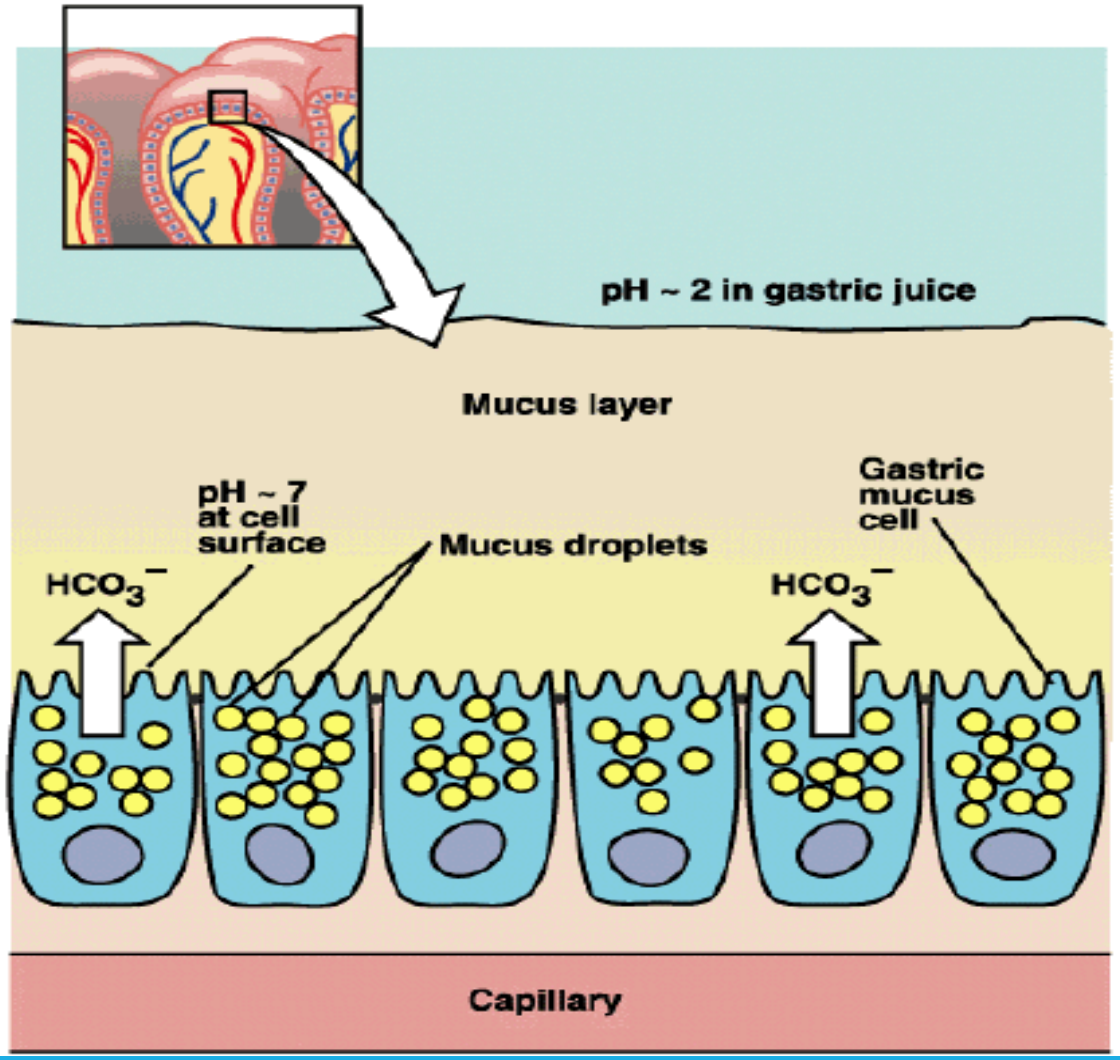
Gastric mucin secretion

- Secreted from surface mucous cells and mucus neck cells
- Glycoprotein
- Functions:
- Barrier to H+
- Combine with HCO3- to form gel
- Protect surface epithelial injury from H+ , Pepsin
- 0.2-0.6 mm. thickness
- Decreased in aging, NSAIDs

Gastric bicarbonate secretion

- Secreted from surface mucous cells
- Through HCO3/Cl exchanger
- Stimulated by: PGE2, vagal stimulation
- Inhibited by: NSAIDs, alpha-adrenergic agonist

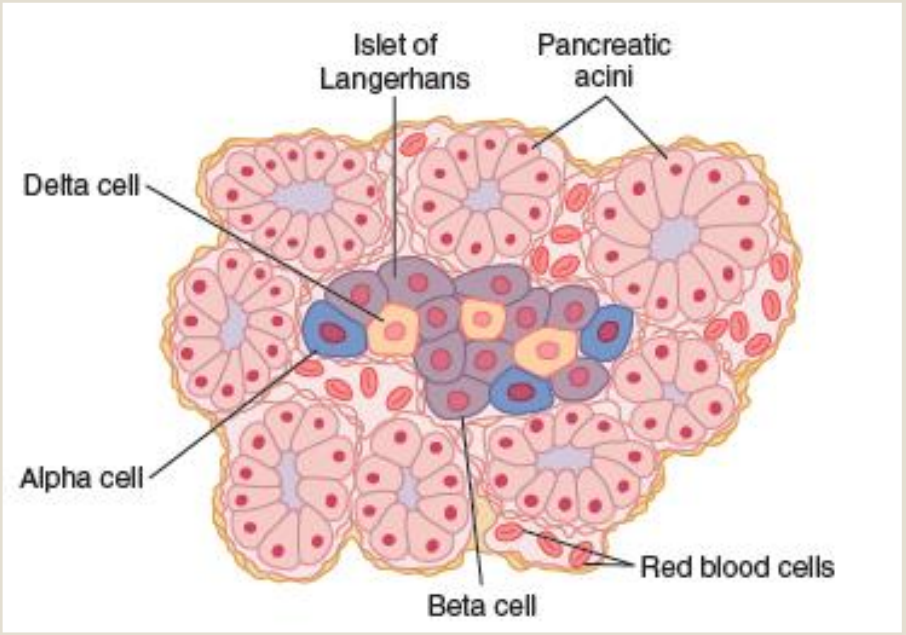
A protective role for mucus

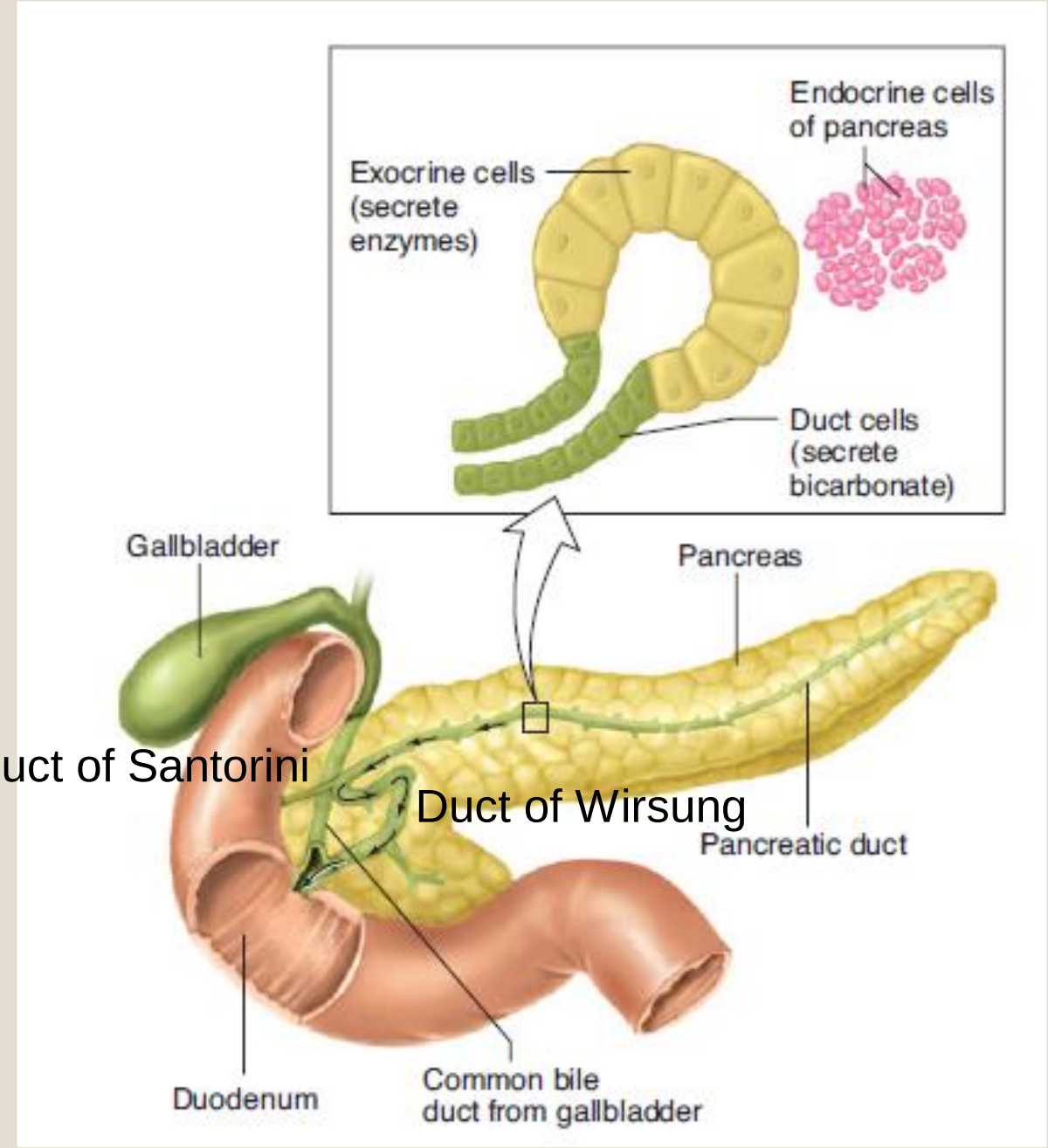


PANCREATIC SECRETION

Pancreatic secretion

- **Exocrine pancreas**: pancreatic enzymes (acini), bicarbonate (ducts) → into duodenum
- **Endocrine pancreas**: insulin, glucagon, somatostatin → into the blood




Endocrine cells
of pancreas
Exocrine cells
(secrete
enzymes)
Duct cells
(secrete
bicarbonate)
Gallbladder
Pancreas
Duct of Santorini
Duct of Wirsung
Pancreatic duct
Duodenum
Common bile
duct from gallbladder

Composition of pancreatic secretion

- Copious amount: 1-1.5 L per day
- Isotonic to plasma
- High bicarbonate concentration (alkaline)
- Low chloride concentration
- Similar Na, K
- Pancreatic enzymes

• BOX 30.1 Products of Pancreatic Acinar Cells

Precursors of Proteases

Trypsinogen
Chymotrypsinogen
Proelastase
Procarboxypeptidase A
Procarboxypeptidase B

Starch-Digesting Enzymes

Amylase

Lipid-Digesting Enzymes or Precursors

Lipase
Nonspecific esterase
Prophospholipase A_2

Nucleases

Deoxyribonuclease
Ribonuclease

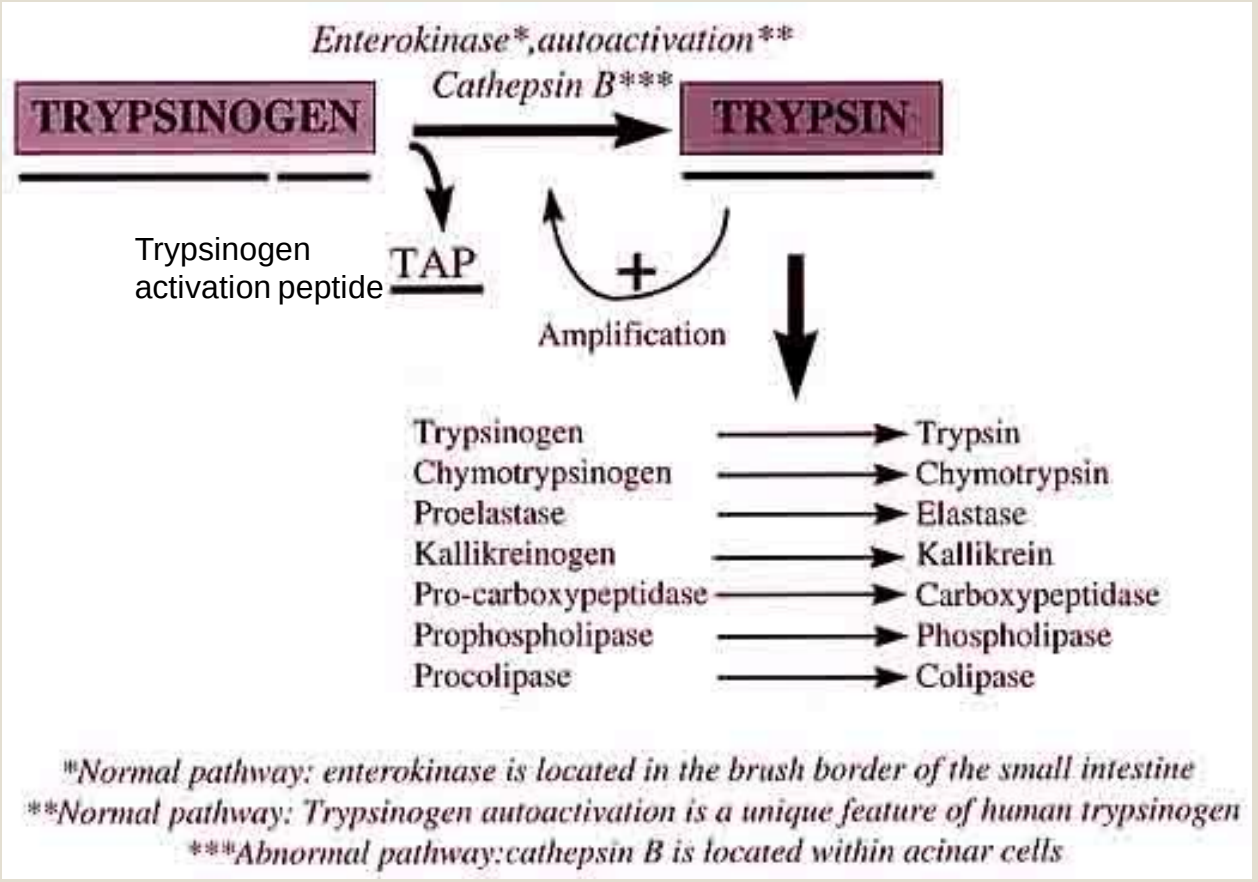
Regulatory Factors

Procolipase
Trypsin inhibitors
Monitor peptide

Pancreatic digestive enzymes

- Proteolytic enzymes are synthesized in inactive forms i.e. trypsinogen, chymotrypsinogen, and procarboxypolypeptidase
- They are activated after secreted into GI tract

Pancreatic enzyme activation



Trypsin inhibitor

- Secreted from the same cells as proteolytic enzymes
- Prevents activation of trypsin both inside the secretory cells and in the acini and ducts of the pancreas
- Genetic mutation of trypsin inhibitor → activation of trypsin and other enzymes inside the pancreas → autodigestion → acute pancreatitis

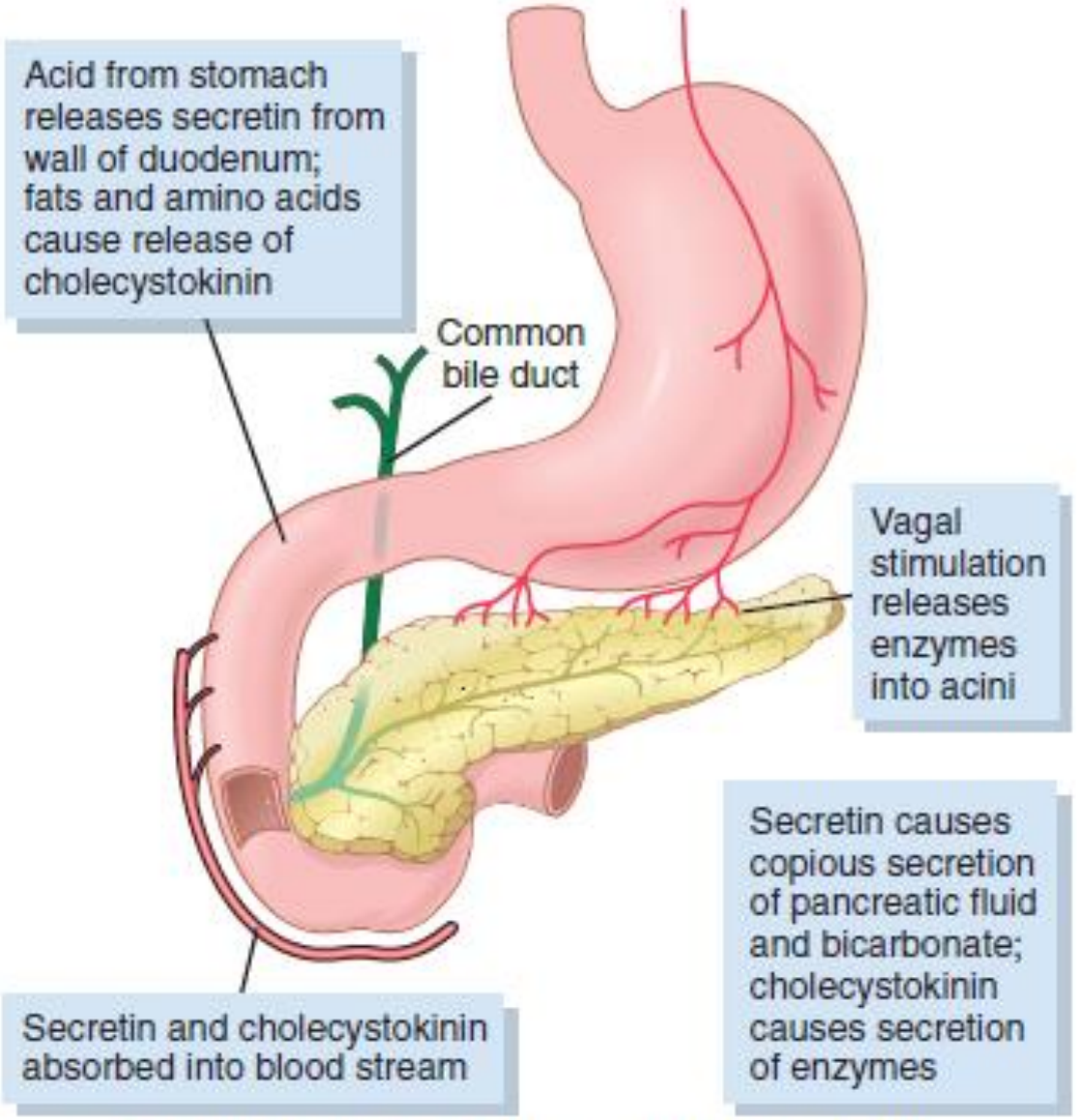


Figure 65-10. Regulation of pancreatic secretion.

INTESTINAL SECRETION

Secretion from Brunner's glands

- Secrete large amounts of alkaline mucus in response
- **Functions:** protect the duodenal wall from digestion by the highly acidic gastric juice, neutralizing HCl
- Inhibited by sympathetic stimulation (stress → ↓ alkaline mucus →↑ risk of duodenal ulcer)

Secretion from Crypts of Lieberkuhn

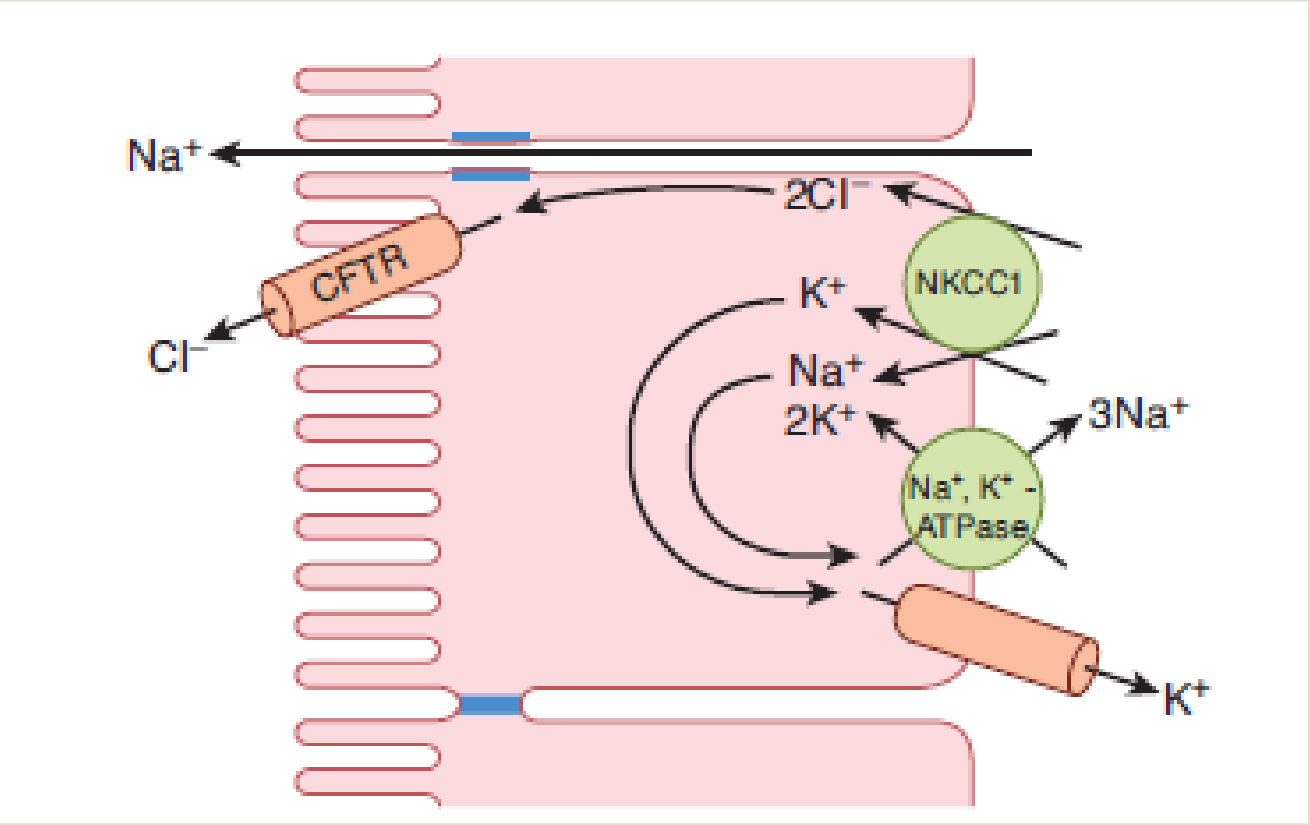
- Located over the entire surface of the small intestine
- 2 types of cells
- **Goblet cells**: secrete mucus that lubricates and protects the intestinal surfaces
- **Enterocytes**: secrete large quantities of water and electrolytes and, over the surfaces of adjacent villi, reabsorb the water and electrolytes along with the end products of digestion

Intestinal secretion

- About 1800 mL/d of fluid secreted from enterocytes
- Slightly alkaline (pH 7.5-8)
- Supplies a watery vehicle for absorption of substances from chyme
- Most of these secretions are rapidly reabsorbed

Mechanism of intestinal secretion

Through chloride and HCO3 secretion



Digestive enzymes of small intestine

- Enterocytes of the mucosa overlying the villi contain digestive enzymes that digest specific food substances **while they are being absorbed** through the epithelium (**not secreted into the lumen**)
- (1) Several peptidases: split small peptides into amino acids
- (2) Sucrase, maltase, isomaltase, and lactase: split disaccharides into monosaccharides
- (3) Small amounts of intestinal lipase: split neutral fats into glycerol and fatty acids.

Regulation of small intestinal secretion

- Most important regulator: local enteric nervous reflexes
- Especially reflexes initiated by tactile or irritative stimuli from the chyme in the intestines.
- Important neurotransmitter of ENS: VIP

COLONIC SECRETION

Colonic secretion

- Crypts of Lieberkühn contain lots of mucous cells that secrete mucus and the remaining cells that secrete bicarbonate
- **Regulation of mucus secretion**
- Direct, tactile stimulation of the epithelial cells lining the large intestine and by local nervous reflexes to the mucous cells (main)
- Stimulation of parasympathetic NS through pelvic n → ↑ mucus secretion

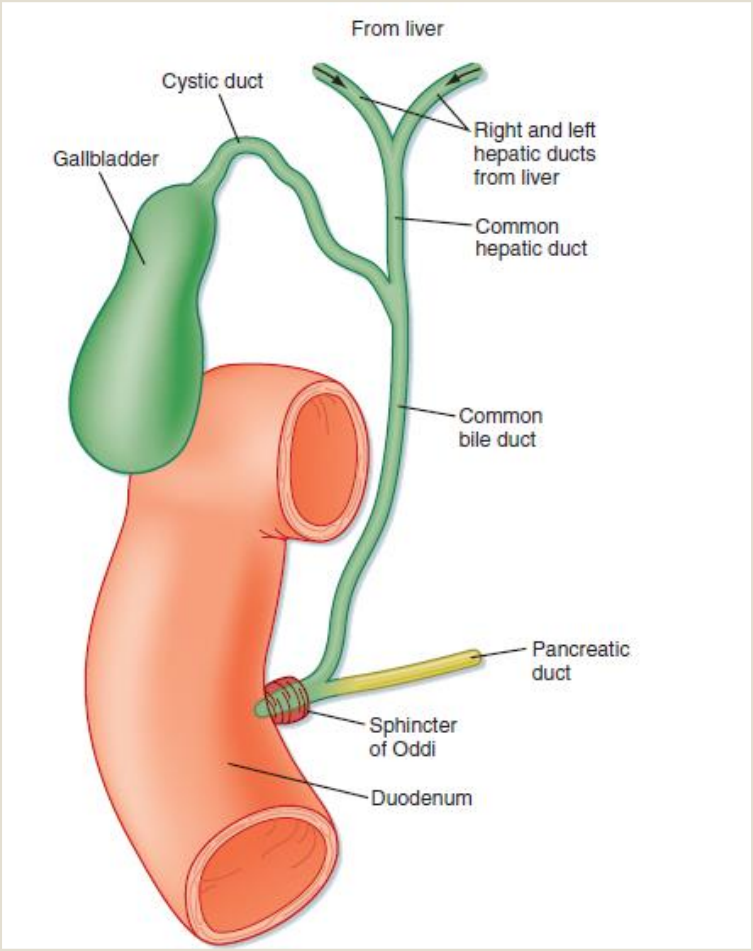
Functions of colonic mucus

- Protects the intestinal wall against excoriation
- Provides an adherent medium for holding fecal matter together.
- Protects the intestinal wall from the great amount of bacterial activity that takes place inside the feces
- Mucus plus the alkalinity of the secretion (a pH of 8.0 caused by large amounts of sodium bicarbonate) provides a barrier to keep acids formed in the feces from attacking the intestinal wall

BILIARY SECRETION

Anatomy of biliary system

- Hepatocytes secrete bile into bile canaliculi → bile ductules (lined with cholangiocytes) → right and left hepatic ducts → common hepatic duct → cystic duct and gallbladder or common bile duct into small intestine



Bile formation and secretion

- Bile is composed of bile acids, phosphatidylcholine, and cholesterol in an approximate ratio of 10:3:1
- Other components: water, electrolytes, bilirubin
- Bile changes its composition as it passes through bile ductules and gallbladder

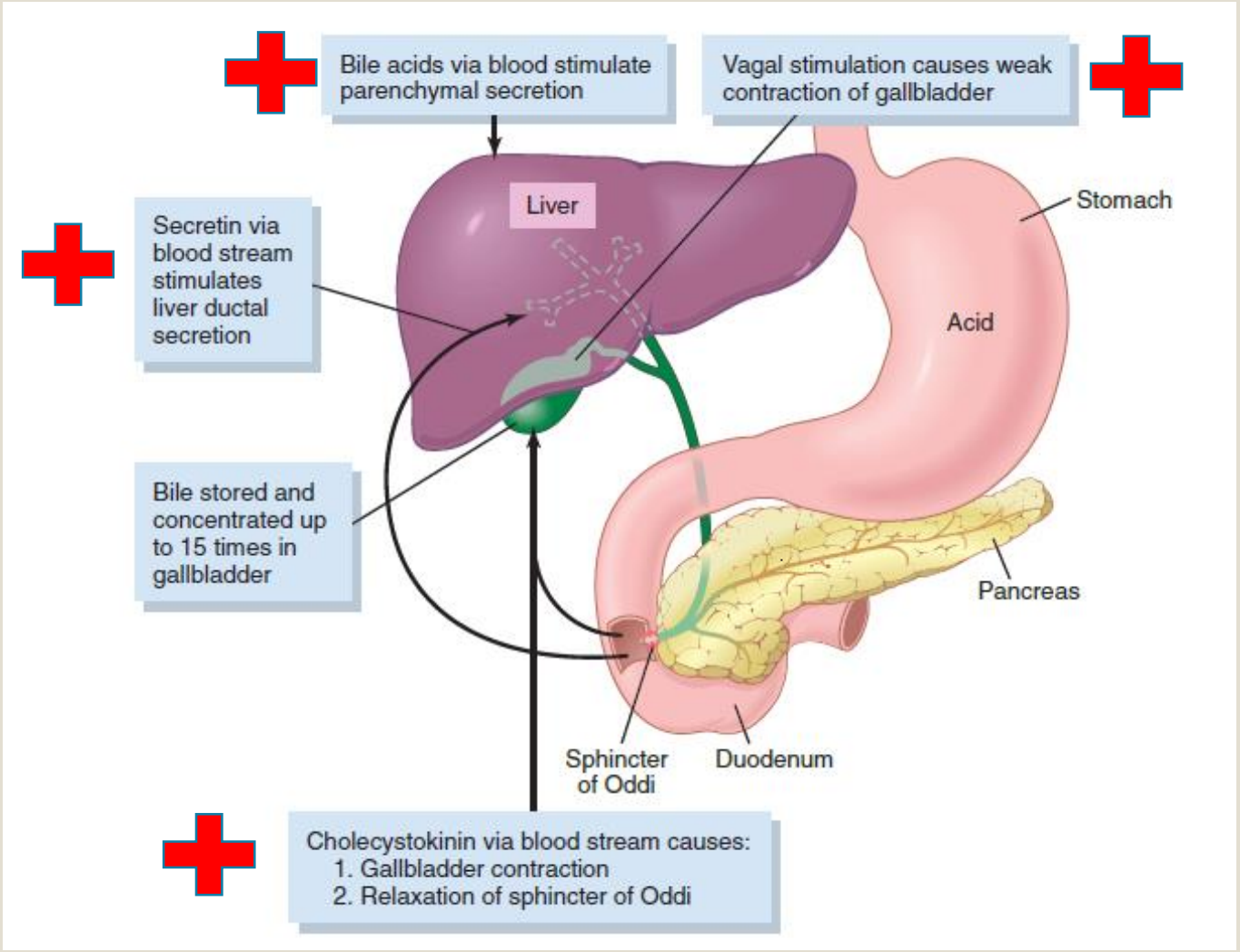
Function of bile

- Fat emulsification and formation of micelles → aid in fat digestion and absorption (bile salt)
- A way to dispose lipid soluble xenobiotics, cholesterol, lipid soluble metabolite end product, bilirubin

Bile acid synthesis

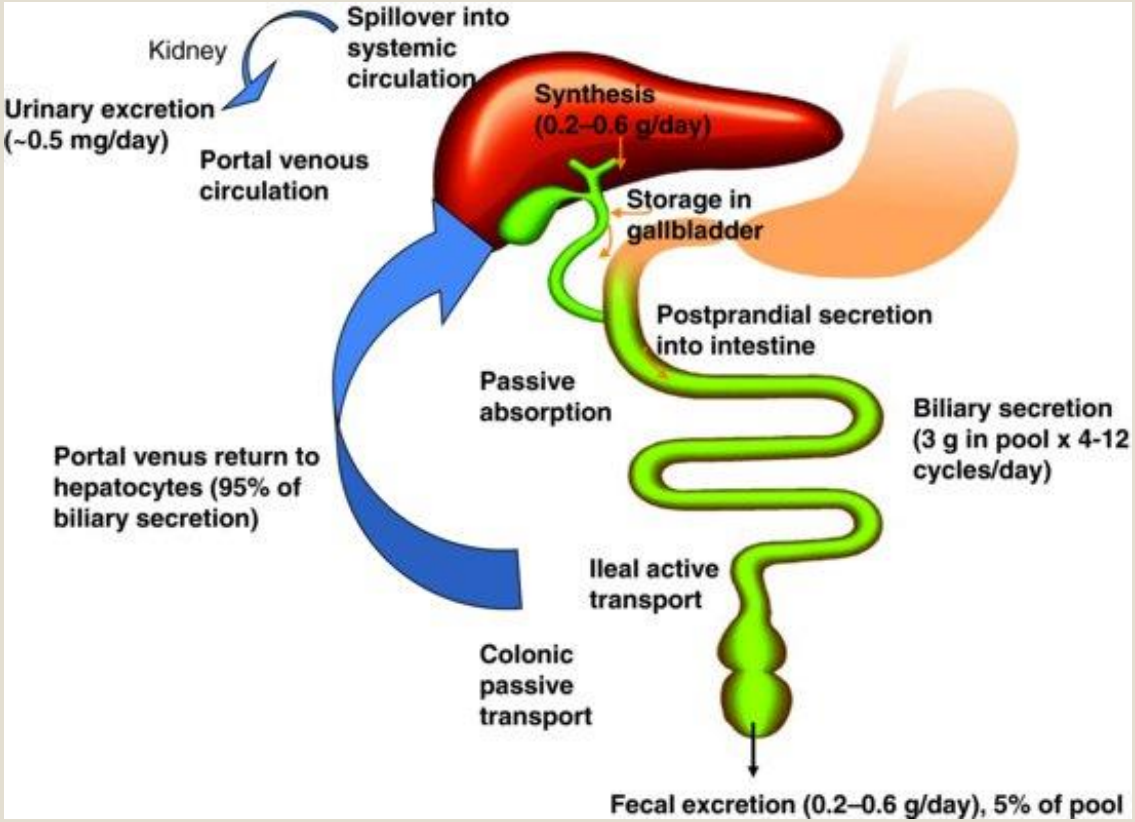
- Bile acid is produced as an end product of cholesterol metabolism
- Rate limiting step enzyme is cholesterol 7α-hydroxylase
- The rate of bile acid synthesis depends on body's requirement (bile acid pool)
- Chenodeoxycholic and cholic acids are primary acids (produced by liver)

Regulation of bile secretion



Enterohepatic circulation

Liver only produces bile acid equal to the amount that lost in the feces each day (200-400 gm/d)



Summary of GI secretion

GI secretion	Characteristic	Stimulated by	Inhibited by
Saliva	High HCO_3^-	Parasympathetic	sleep
	High K^+	Sympathetic	dehydration
	Hypotonic		atropine
	Amylase, lipase		
Gastric	HCl	Gastrin	↓ stomach pH
		Parasympathetic	chyme in duodenum
		Histamine	atropine , cimetidine
			omeprazole
	Pepsinogen	Ach, gastrin	
	Intrinsic factor		

Summary of GI secretion

GI secretion	Characteristic	Stimulated by	Inhibited by
Pancreatic	High HCO_3^- Isotonic Pancreatic lipase, amylase , protease	Secretin Parasympathetic CCK	
Bile	Bile salt Bilirubin Phospholipid , cholesterol	bile salt Parasympathetic secretin CCK	Ileal resection

THE END

Thank you for your attention